

DP

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1 Introduction

Let A be a length n array of integers. Find the length of a longest increasing subsequence.

Existence of solution. A subsequence is a nonempty subset of $[n]$, which is finite, so the set of increasing subsequences is finite. Hence, there exists a longest increasing subsequence.

Subproblem. Let $\text{OPT}[i]$ denote the length of a longest increasing subsequence ending at i for $i \in [n]$.

Recurrence. Base case. Each $i \in [n]$ has a length-one increasing subsequence ending at i : $\{i\}$. Inductive step. Let $X = \{i_1, i_2, \dots, i_m\}$ be a longest increasing subsequence ending at $i_m = k + 1$. If $m = 1$, $\text{OPT}[k + 1] = 1$. Else, $X' = X - \{k + 1\} \neq \emptyset$ is a longest increasing subsequence ending at some $i_{m-1} \leq k$, so $\text{OPT}[k + 1] = 1 + \text{OPT}[k']$ for some $k' \leq k$. Conversely, if $X' = \{i_1, i_2, \dots, i_m\}$